

INTERNSPARK MERN STACK INTERNSHIP

TASK 2: Frontend SPA with React (Fetch API & Routing)

Project Name:

InternHub – Internship Management Portal

Submitted By:

MD Sohail Ahmed

Technology Stack:

React

React Router DOM

Axios

Tailwind CSS

Vite

Node.js Backend API

MongoDB Atlas

Submission Date: 02/06/2026

Objective:

The objective of this task is to develop a React-based Single Page Application (SPA) that consumes the Internship Management REST API developed in Task 1. The application provides complete CRUD functionality, client-side routing, API integration, loading and error handling, and a responsive user interface for managing internship records.

The frontend communicates with the backend API using Axios and dynamically displays internship information stored in MongoDB Atlas.

Technologies Used:

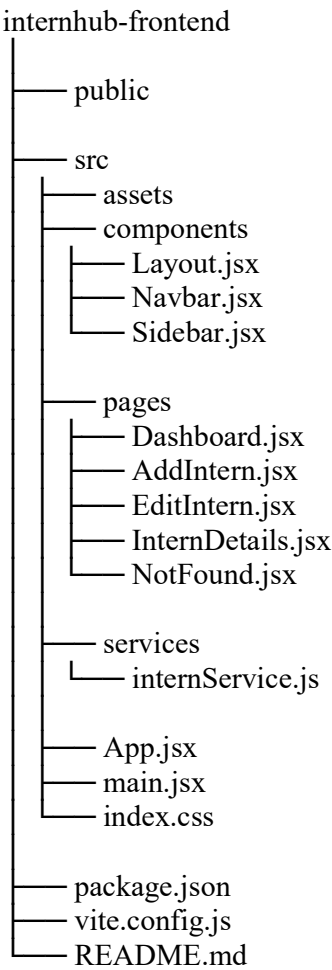
The following technologies were used to develop the InternHub Frontend Application:

1. React.js
 - Used for building reusable user interface components.
 - Implemented using functional components and React Hooks.
2. React Router DOM
 - Used for client-side routing and navigation between pages.
3. Axios
 - Used for communicating with the backend REST API.
4. Tailwind CSS
 - Used for responsive and modern UI styling.
5. Vite
 - Used as the frontend build tool and development server.
6. Node.js & Express.js API
 - Backend service developed in Task 1.
7. MongoDB Atlas
 - Cloud database used for storing internship records.

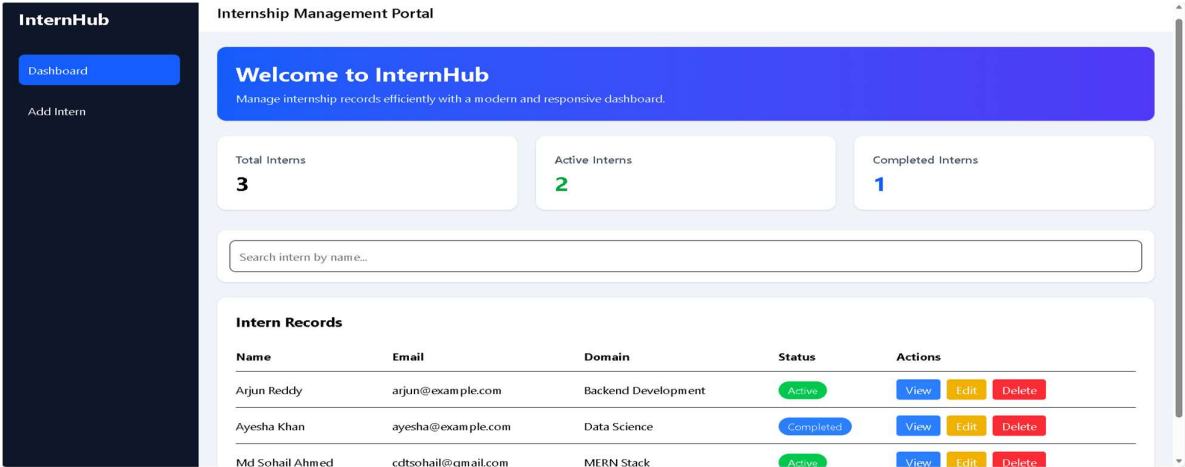
Project Structure:

The frontend application follows a modular structure to improve maintainability and scalability.

Folder Structure:



Dashboard Overview:



The Dashboard serves as the central hub of the InternHub application. It displays internship statistics, provides a searchable internship table, and offers quick access to CRUD operations through action buttons. The dashboard fetches internship records dynamically from the Express.js backend API using Axios.

Key Features:

- Internship Statistics Cards
- Search Functionality
- Responsive Data Table
- View, Edit and Delete Actions
- Active Navigation Sidebar
- Welcome Banner

Create Intern Functionality:

InternHub

Dashboard

Add Intern

Internship Management Portal

Add New Intern

Md Sohail Ahmed

cdtsohail@gmail.com

9912365455

Anurag University

MERN Stack

Active

06/01/2026

Notes

InternHub

Dashboard

Add Intern

Internship Management Portal

Dashboard

Total Interns

4

Active Interns

3

Completed

1

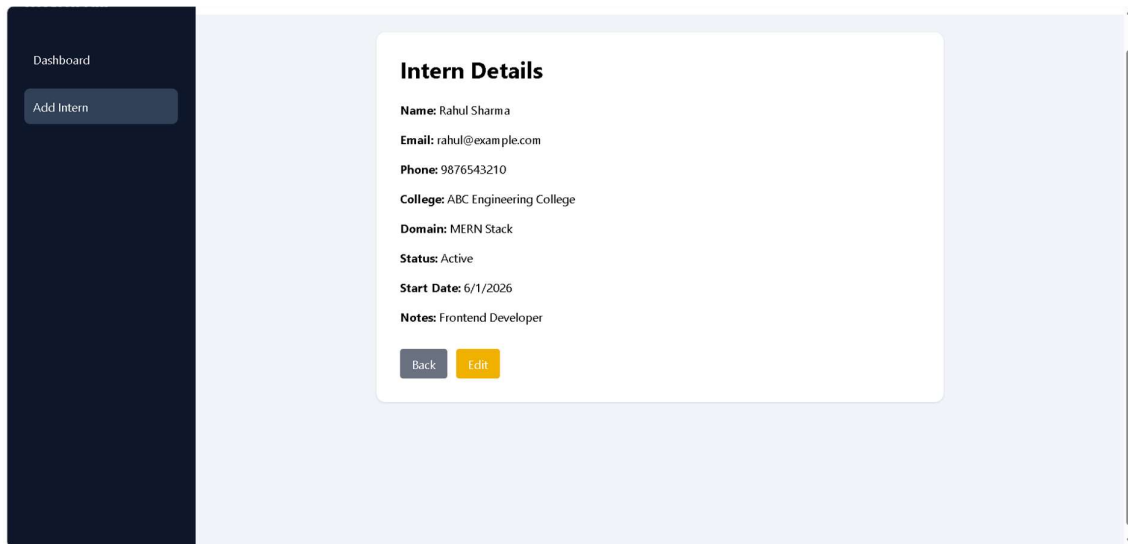
Intern Records

Name	Email	Domain	Status
Rahul Sharma	rahul@example.com	MERN Stack	Active
Arjun Reddy	arjun@example.com	Backend Development	Active
Ayesha Khan	ayasha@example.com	Data Science	Completed
Md Sohail Ahmed	cdtsohail@gmail.com	MERN Stack	Active

The Create Intern page allows users to add new internship records into the system. The form collects internship details such as name, email, phone number, college, domain, status, start date, and notes.

Upon submission, the frontend sends a POST request to the backend API using Axios and stores the data in MongoDB Atlas.

View Intern Details:



Intern Details

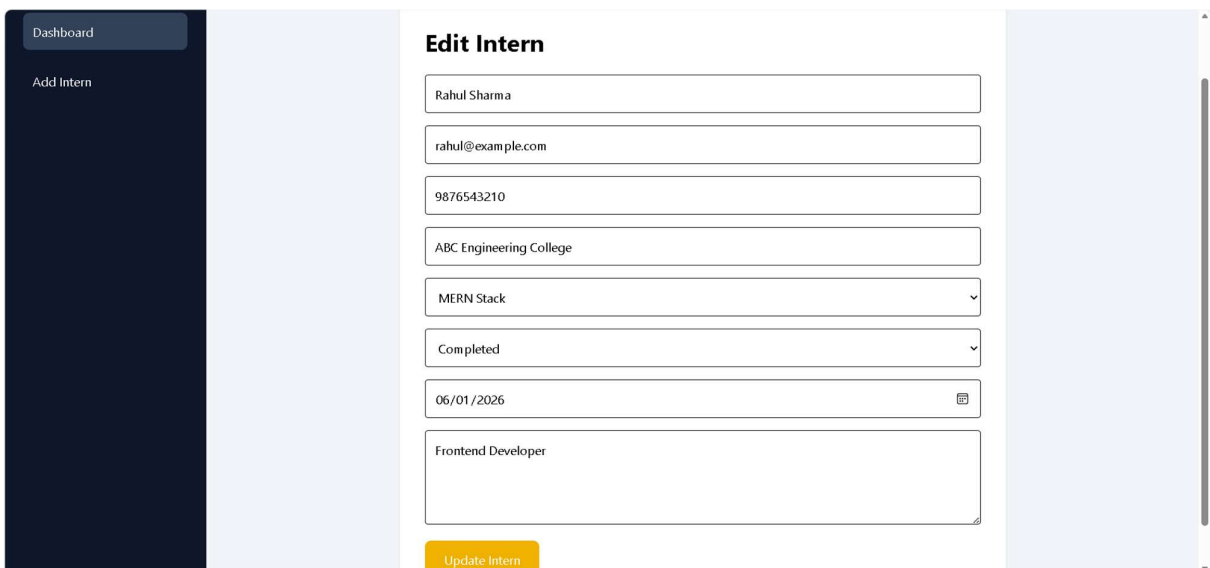
Name: Rahul Sharma
Email: rahul@example.com
Phone: 9876543210
College: ABC Engineering College
Domain: MERN Stack
Status: Active
Start Date: 6/1/2026
Notes: Frontend Developer

[Back](#) [Edit](#)

The View Intern Details page retrieves and displays complete information about a selected intern. Dynamic routing is implemented using React Router to load internship information based on the intern ID.

This functionality demonstrates successful communication between React, Express.js, and MongoDB Atlas.

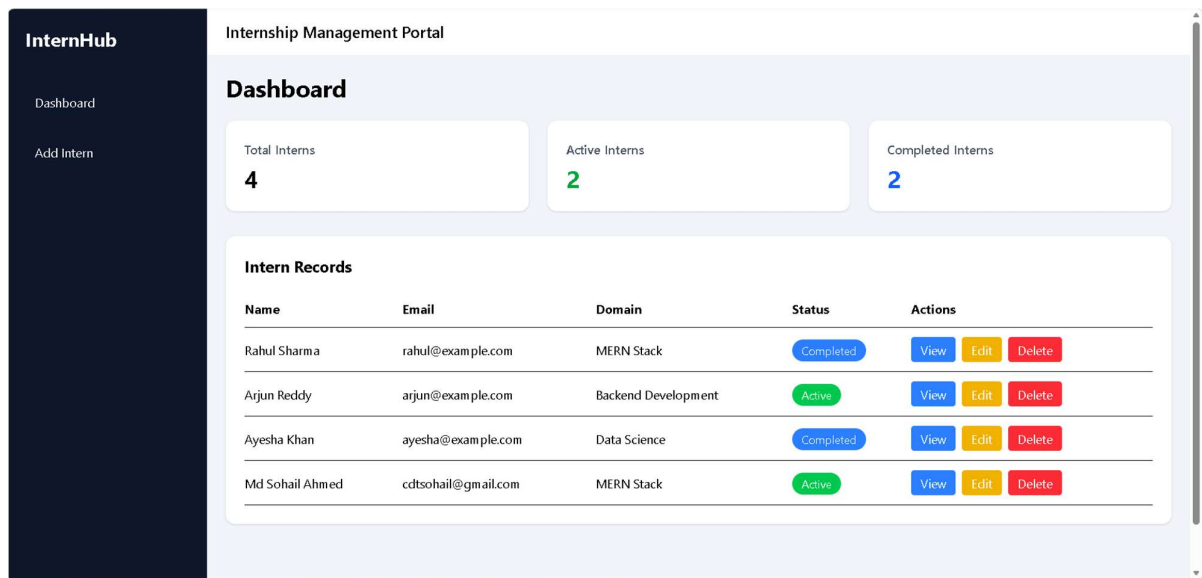
Update Intern Records:



Edit Intern

Rahul Sharma
rahul@example.com
9876543210
ABC Engineering College
MERN Stack
Completed
06/01/2026
Frontend Developer

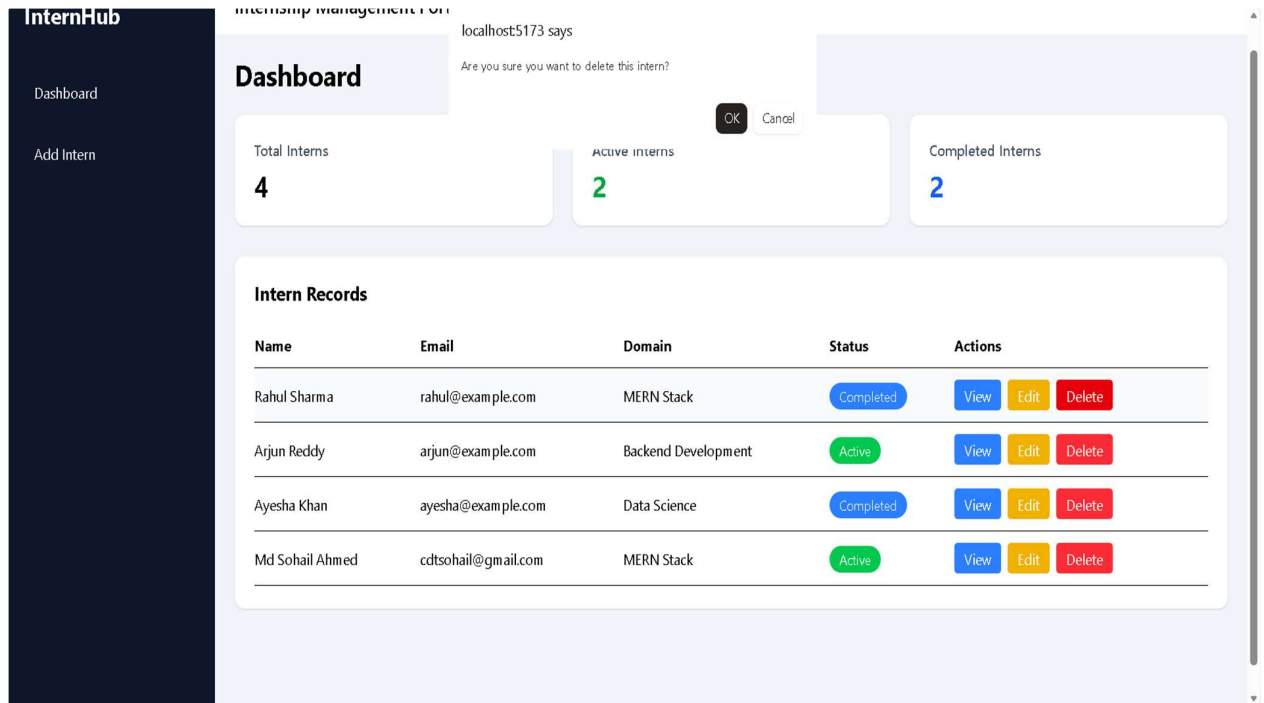
[Update Intern](#)

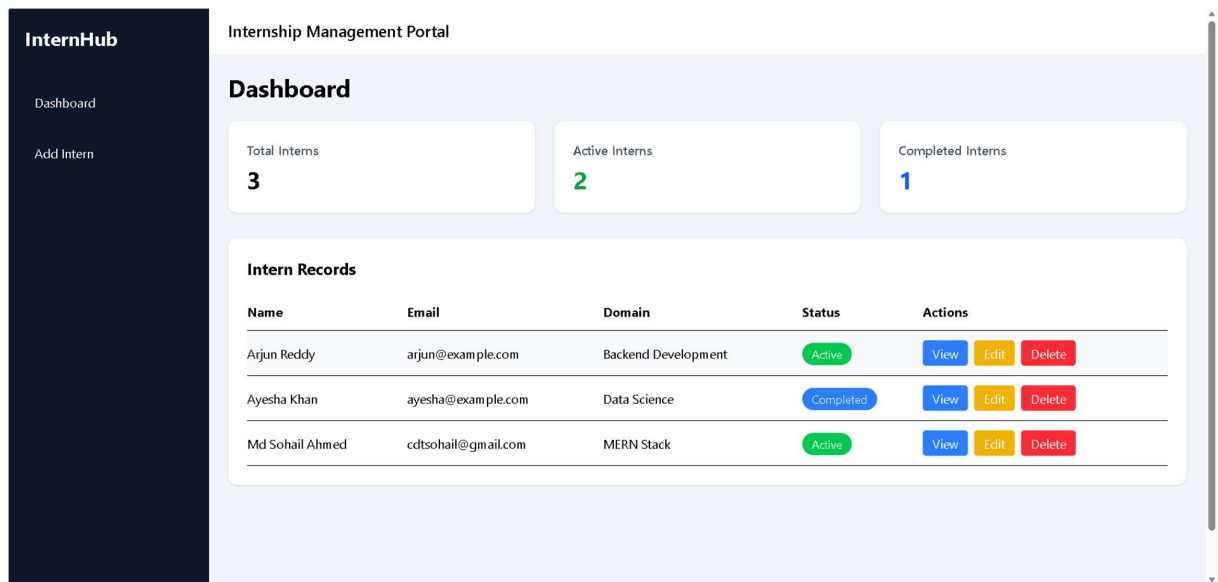


The Edit Intern page enables modification of existing internship records. Existing data is fetched automatically and pre-filled into the form. Updated information is submitted through an Axios PUT request and saved in MongoDB Atlas.

The dashboard statistics and internship records are updated accordingly.

Delete Intern Records:

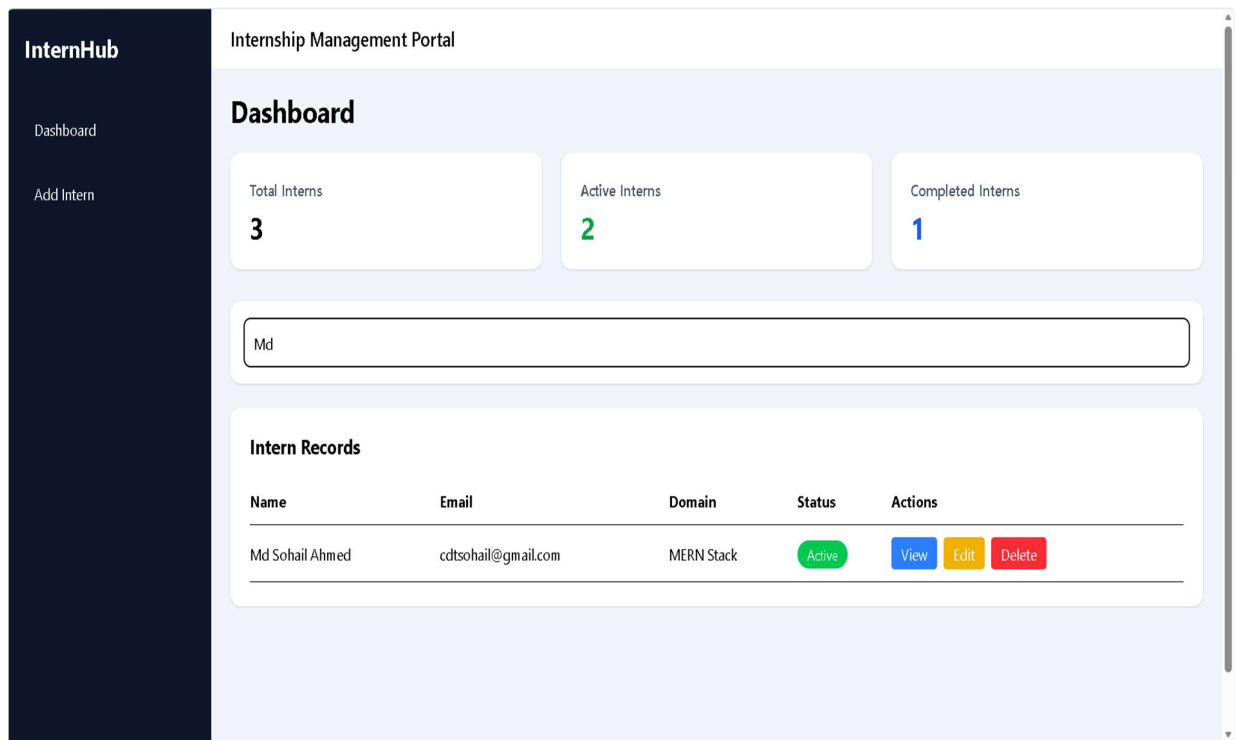


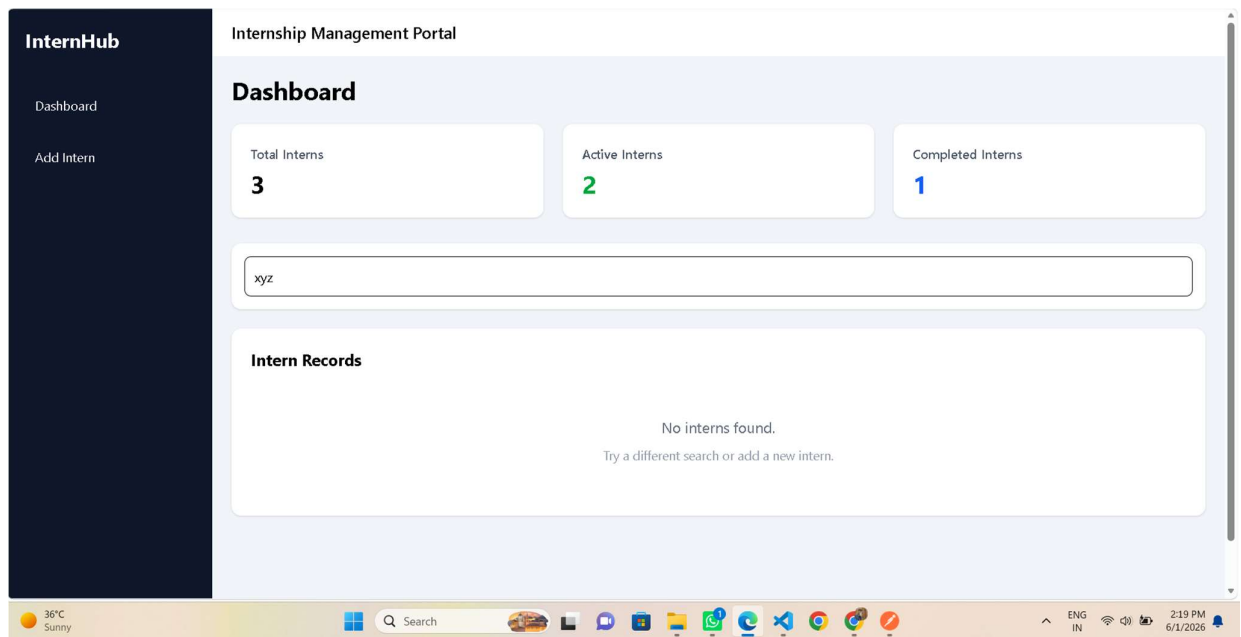


The Delete functionality allows users to permanently remove internship records. A confirmation dialog is displayed before deletion to prevent accidental data loss.

Upon confirmation, an Axios DELETE request is sent to the backend API, and the dashboard refreshes automatically to display updated data.

Search, Loading and Error Handling:





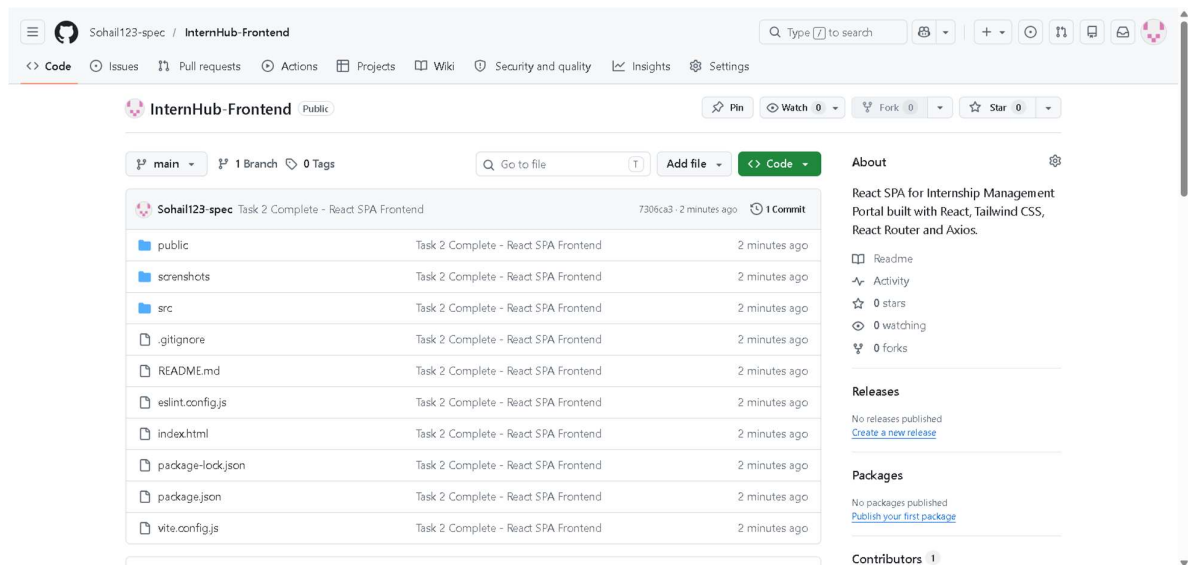
Additional user experience enhancements were implemented to improve application usability.

Features Implemented:

- Search Interns by Name
- Loading State while fetching records
- Error Handling for API failures
- Empty State when no records are found

These features improve responsiveness and provide clear feedback to users during interactions.

GitHub Repository:



GitHub Repository Link:

<https://github.com/Sohail123-spec/InternHub-Frontend>

The complete source code for the React frontend application has been uploaded to GitHub for version control, collaboration, and project documentation purposes.

Deliverables:

- ✓ **React SPA built using React and Vite**
- ✓ **CRUD operations implemented**
- ✓ **React Router Navigation**
- ✓ **Axios API Integration**
- ✓ **Loading and Error Handling**
- ✓ **GitHub Repository**
- ✓ **Project Documentation**

Conclusion:

The InternHub Frontend SPA was successfully developed using React, React Router DOM, Axios, and Tailwind CSS. The application integrates seamlessly with the backend REST API developed in Task 1 and provides complete CRUD functionality through a responsive and user-friendly interface.

Additional features such as search functionality, loading states, error handling, active navigation, and dashboard analytics were implemented to enhance usability and user experience. The project successfully fulfills all requirements specified in Task 2.